study_id	study	device	doi	registry_id	first_publication_date	design	publication_cohort	follow_up	primary_endpoint	principal_result	lead_sponsor	funding	oversight_or_analysis	independence_class	ai_effect_isolated	applicability	limitations	access_level	source_url	registry_url	abstract_record	accessed_date
T02	ATLAS	EMBLEM S-ICD vs transvenous ICD	10.7326/M22-1566	NCT02881255	2022-11-08	Investigator-sponsored randomized multicenter open-label parallel group	544 eligible screened; 503 randomized (251 / 252)	Primary endpoint six months; mean follow-up 2.5 years	Major perioperative lead-related complications	1/251 (0.4%) vs 12/252 (4.8%); risk difference −4.4 percentage points (95% CI −6.9 to −1.9); journal P=0.001	Population Health Research Institute	Boston Scientific	PHRI trial; expert committee central adjudication and DSMB described in investigator slides; manufacturer role in all analyses not established from abstract/slides	Academic-led / industry-funded	No	Directly supports component-level lead complication claims, not AI or combined EMPOWER therapy	Underpowered for shock outcomes; early pain greater with S-ICD; inappropriate-shock HR 2.37 (95% CI 0.98–5.77); cohort selected for age/risk	Primary journal abstract via Europe PMC; primary investigator slides and PHRI site read; journal full text HTTP 403	https://doi.org/10.7326/M22-1566	https://clinicaltrials.gov/study/NCT02881255	https://europepmc.org/article/MED/36343346	2026-10-28
T01	PRAETORIAN	Subcutaneous ICD vs transvenous ICD	10.1056/NEJMoa1915932	NCT01296022	2020-08-06	Investigator-initiated randomized noninferiority; modified intention-to-treat	876 randomized; 849 analyzed (426 S-ICD / 423 TV-ICD)	Median 49.1 months	Device-related complications or inappropriate shocks	48-month cumulative incidence 15.1% vs 15.7%; HR 0.99 (95% CI 0.71–1.39); noninferiority P=0.01; superiority P=0.95	Academic Medical Center Amsterdam	Boston Scientific	Investigators designed and analyzed; company reported no role in design/analysis/manuscript; independent DSMB and clinical-events committee	Academic-led / industry-funded	No	Supports S-ICD component and lead-avoidance rationale, not combined EMPOWER system or AI	Excludes pacing indication; not an EMPOWER trial; 27 post-randomization exclusions; no established superiority for primary composite	Full primary article read from public hospital-hosted PDF; copyrighted, not verified open-licensed	https://www.valleyhealth.com/sites/default/files/2020-09/PRAETORIAN%20NEJM%202020.pdf	https://clinicaltrials.gov/study/NCT01296022	https://europepmc.org/article/MED/32757521	2026-10-28
T03	Qinming8631 DR noninferiority trial	Qinming8631 DR vs Talos DR	10.4103/0366-6999.193443		2016-11-20	Prospective randomized multicenter single-blind noninferiority	225 randomized (113 / 112)	Six months	Effective pacing rate	FAS: 81.4% vs 79.5%; PPS: 95.4% vs 89.5%; authors report noninferiority with predefined −9.5% limit	Institutional clinical investigators; formal registry sponsor not verified	Article reports financial support and sponsorship: Nil	Article describes third-party data processing/statistical analyst; no conflicts declared; external sponsor-control audit unavailable	No sponsorship reported / independence not fully verified	No	Direct Qinming pacing comparison; AI integration not established; QM7211 is a ventricular lead	87/113 and 86/112 in per-protocol completion set; six-month duration; do not treat noninferiority as superiority	Original full article read in existing corpus and verified abstract; publisher marks Open, precise licence unresolved	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx		https://europepmc.org/article/MED/27823996	2026-10-28
T04	MARVEL 2	Micra accelerometer-based AV algorithm; predecessor evidence	10.1016/j.jacep.2019.10.017	NCT03752151	2019-11-11	Prospective acute within-patient VDD vs VVI-50 comparison; not randomized parallel group	75 with algorithm download; efficacy subset 40; current registry enrollment 77	Up to five hours	Proportion with ≥70% AV synchrony at rest	95% with VDD vs 0% with VVI (P<0.001); mean AV synchrony 89.2% vs 26.8%	Medtronic Cardiac Rhythm and Heart Failure	Industry-sponsored; exact funding language in full article not separately audited	Registry notes masked echo core lab; sponsor is Medtronic	Manufacturer-sponsored comparator evidence	No	Tests sensor-driven algorithm performance; not ML validation; not a direct Micra AV2/VR2 trial	Acute surrogate endpoint; predecessor device/software; no long-term outcome comparison	Primary article abstract via Europe PMC plus registry; full article not reviewed	https://doi.org/10.1016/j.jacep.2019.10.017	https://clinicaltrials.gov/study/NCT03752151	https://europepmc.org/article/MED/31709982	2026-10-28
T05	AccelAV	Micra AV, not Micra AV2/VR2	10.1016/j.hrthm.2022.08.033	NCT04245345	2022-09-06	Prospective multicenter single-arm	152 enrolled and implanted; primary subset 54; optimization subset 20; current registry enrollment 157	Primary endpoint one month; follow-up through three months	Resting AV synchrony in complete AV block and normal sinus function	Primary subset mean resting AV synchrony 85.4%; ambulatory 74.8%; optimization subset ambulatory 82.6%	Medtronic Cardiac Rhythm and Heart Failure	Medtronic (registry sponsor and indexed grant information)	Independent sponsor control of analysis not established from abstract	Manufacturer-sponsored comparator evidence	No	Clinical follow-up of mechanical atrial sensing; no AI-specific contrast; generation mismatch to AV2/VR2	Single arm; selected subsets; optimization percentage not whole-cohort result; quality-of-life change not causal comparative effect	Primary abstract plus registry; publisher full text HTTP 403	https://doi.org/10.1016/j.hrthm.2022.08.033	https://clinicaltrials.gov/study/NCT04245345	https://europepmc.org/article/MED/36075532	2026-10-28
T06	AVEIR DR i2i pivotal trial	AVEIR DR dual-chamber leadless pacing	10.1056/NEJMoa2300080	NCT05252702	2023-05-20	Prospective multicenter single-group against performance goals	300 in 2023 paper; current registry enrollment 464	90 days / three months	Freedom from serious device/procedure complications; electrical performance; ≥70% AV synchrony	Implant success 295/300 (98.3%); safety 271/300 (90.3%, 95% CI 87.0–93.7); 29 patients had 35 serious events; AV synchrony criterion met by 97.3%	Abbott Medical Devices	Abbott Medical	Separate independent adjudication/analysis not verified from abstract; industry sponsor established	Manufacturer-sponsored comparator evidence	No	Direct AVEIR DR evidence; communication/pacing performance, not AI benefit	Single group; 97.3% is patients achieving ≥70% synchrony, not average synchrony; 98.3% implant success is not 90-day safety	Primary abstract via Europe PMC plus registry; journal full text not reviewed	https://doi.org/10.1056/NEJMoa2300080	https://clinicaltrials.gov/study/NCT05252702	https://europepmc.org/article/MED/37212442	2026-10-28
T07	MODULAR ATP	EMBLEM S-ICD + EMPOWER	10.1056/NEJMoa2401807	NCT04798768	2024-05-18	Prospective multinational single-group against performance goals	293 enrolled; six-month endpoint cohort 162; 151 completed six-month visit; current registry 297	Six-month primary endpoints	Freedom from major leadless-pacemaker complications; communication success; pacing threshold	97.5% free from major pacemaker complications; 98.8% of communication tests successful; 147/151 (97.4%) acceptable pacing threshold	Boston Scientific Corporation	Boston Scientific	Independent monitoring and clinical-events committees; sponsor collected/monitored data and performed outcome analyses	Manufacturer-sponsored / independent adjudication	No	Direct combined-system clinical evidence; no AI-specific contrast; later update only	Published after February 2024 paper; performance goals not concurrent controls; endpoint cohort smaller than enrollment	Full primary article publicly reachable via clinical mirror; copyrighted, no open reuse licence verified	https://www.cardioaragon.com/wp-content/uploads/A-Modular-Communicative-Leadless.NEJMed.2024.pdf	https://clinicaltrials.gov/study/NCT04798768	https://europepmc.org/article/MED/38767244	2026-10-28

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	T01	PRAETORIAN	We conducted this investigator-initiated, international, randomized, noninferiority trial in the United States and Europe.	Article p.527, Trial Design and Oversight	GOVERNANCE	Academic initiation, not industry-free funding.	https://www.valleyhealth.com/sites/default/files/2020-09/PRAETORIAN%20NEJM%202020.pdf	10.1056/NEJMoa1915932	2020-08-06	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C02	T01	PRAETORIAN	This trial was funded by Boston Scientific, which had no role in the design of the trial, analysis of the data, or the drafting and submission of the manuscript.	Article p.527, Trial Design and Oversight	FUNDING_ROLE	Industry funding with reported investigator control.	https://www.valleyhealth.com/sites/default/files/2020-09/PRAETORIAN%20NEJM%202020.pdf	10.1056/NEJMoa1915932	2020-08-06	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C03	T01	PRAETORIAN	A clinical-events committee consisting of three electrophysiologists who were not otherwise involved in the trial adjudicated all clinical and arrhythmic events.	Article p.528, End Points	ADJUDICATION	Independent event review is a separate dimension from funding.	https://www.valleyhealth.com/sites/default/files/2020-09/PRAETORIAN%20NEJM%202020.pdf	10.1056/NEJMoa1915932	2020-08-06	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C04	T02	ATLAS	Investigator Sponsored Trial with funding from Boston Scientific	HRS 2022 investigator presentation, slide 5	FUNDING_ROLE	Academic sponsorship does not mean no manufacturer funding.	https://www.phri.ca/wp-content/uploads/2022/04/ATLAS-S-ICD_HRS2022-1.pdf	10.7326/M22-1566	2022-11-08	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C05	T02	ATLAS	which occurred in 1 patient (0.4%) with an S-ICD and in 12 patients (4.8%) with TV-ICD	Primary journal abstract > Results	CLINICAL_OUTCOME	Lead-related complications; not an AI effect.	https://doi.org/10.7326/M22-1566	10.7326/M22-1566	2022-11-08	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C06	T02	ATLAS	At present, the ATLAS trial is underpowered to detect differences in clinical shock outcomes; however, extended follow-up is ongoing.	Primary journal abstract > Limitation	INFERENCE_LIMIT	Nonsignificant shock differences do not establish equivalent shock outcomes.	https://doi.org/10.7326/M22-1566	10.7326/M22-1566	2022-11-08	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C07	T03	Qinming8631 DR noninferiority trial	Nil.	Original article > Financial support and sponsorship	FUNDING_ROLE	Self-reported absence of sponsorship, not external verification of independence.	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx	10.4103/0366-6999.193443	2016-11-20	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C08	T03	Qinming8631 DR noninferiority trial	Data were collected at each clinical site by a third-party analyzer who was responsible for data processing and statistical analysis.	Original article > Methods > Study design	ANALYSIS_ROLE	Reported third-party processing; identity and contract independence not established.	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx	10.4103/0366-6999.193443	2016-11-20	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C09	T04	MARVEL 2	downloaded to the Micra leadless pacemaker for up to 5 h in patients with AV block.	Primary abstract > Methods	FOLLOWUP_LIMIT	Acute algorithm assessment, not a chronic AI clinical trial.	https://doi.org/10.1016/j.jacep.2019.10.017	10.1016/j.jacep.2019.10.017	2019-11-11	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C10	T04	MARVEL 2	Among the 40 patients with sinus rhythm and complete AV block included in the primary efficacy objective analysis	Primary abstract > Results	DENOMINATOR	Efficacy subset is not all 75 participants.	https://doi.org/10.1016/j.jacep.2019.10.017	10.1016/j.jacep.2019.10.017	2019-11-11	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C11	T05	AccelAV	AccelAV was a prospective, single-arm study to characterize AVS in patients implanted with a Micra AV	Primary abstract > Methods	TRIAL_DESIGN	No randomized concurrent control; Micra AV is a predecessor generation.	https://doi.org/10.1016/j.hrthm.2022.08.033	10.1016/j.hrthm.2022.08.033	2022-09-06	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C12	T05	AccelAV	Among patients with normal sinus function and complete AVB (n = 54), mean resting AVS was 85.4% at 1 month, and ambulatory AVS was 74.8%.	Primary abstract > Results	DENOMINATOR	Report resting and ambulatory values separately; subset n=54.	https://doi.org/10.1016/j.hrthm.2022.08.033	10.1016/j.hrthm.2022.08.033	2022-09-06	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C13	T06	AVEIR DR i2i pivotal trial	We conducted a prospective, multicenter, single-group study to evaluate the safety and performance of a dual-chamber leadless pacemaker system.	Primary abstract > Methods	TRIAL_DESIGN	Performance-goal comparison, not randomized superiority.	https://doi.org/10.1056/NEJMoa2300080	10.1056/NEJMoa2300080	2023-05-20	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C14	T06	AVEIR DR i2i pivotal trial	A total of 35 device- or procedure-related serious adverse events occurred in 29 patients.	Primary abstract > Results	SAFETY	Adds harms omitted from simplified implant-success reporting.	https://doi.org/10.1056/NEJMoa2300080	10.1056/NEJMoa2300080	2023-05-20	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C15	T06	AVEIR DR i2i pivotal trial	At least 70% atrioventricular synchrony was achieved in 97.3% of the patients	Primary abstract > Results	DENOMINATOR	97.3% is proportion meeting a threshold, not a mean synchrony measure.	https://doi.org/10.1056/NEJMoa2300080	10.1056/NEJMoa2300080	2023-05-20	2026-10-28	PASS: exact primary-abstract substring after whitespace normalization; retrieved via Europe PMC
C16	T07	MODULAR ATP	An independent clinical events committee reviewed adverse events and episodes of arrhythmia that were treated.	Article p.1403, Study Design and Oversight	ADJUDICATION	Independent adjudication present despite manufacturer sponsorship.	https://www.cardioaragon.com/wp-content/uploads/A-Modular-Communicative-Leadless.NEJMed.2024.pdf	10.1056/NEJMoa2401807	2024-05-18	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only
C17	T07	MODULAR ATP	The sponsor collected and monitored study data and performed outcome analyses according to the statistical analysis plan and the protocol	Article p.1403, Study Design and Oversight	ANALYSIS_ROLE	Independent review does not mean sponsor-independent analysis.	https://www.cardioaragon.com/wp-content/uploads/A-Modular-Communicative-Leadless.NEJMed.2024.pdf	10.1056/NEJMoa2401807	2024-05-18	2026-10-28	Read primary full text/slide or primary abstract; PDF line-wrap hyphenation and whitespace normalized only

device	verification_verdict	mechanism	processing_location	training_or_model_evidence	validation_scope	evidence_ids	confidence_and_limits	safe_description
	AI-assisted AF classification verified; architecture unresolved	Uploaded subcutaneous ECG → QRS/signal processing and AF AI classification → true/false episode label → clinician review	Home Monitoring Service Center, not implant or programmer	Patient ECG training explicitly reported; evaluation excludes training contributors; model architecture, weights and complete training data not disclosed	368 patients; 61,517 sECGs; balanced setting; observational/replayed workflow with IIIm recordings approximating IV	V01; M01–M05	High for location and AF scope; insufficient for architecture or causal clinical benefit	AI-assisted remote ECG review associated with an implanted monitor; not implant-resident adaptive therapy
Micra AV2	Automatic sensor-driven control verified; AI/ML not established	Mechanical atrial activity sensed by accelerometer; A3/A4 threshold setup; Auto+ adjusts A3 threshold; Auto PVAB adjusts blanking by ventricular rate	Implanted ventricular pacemaker	No trained statistical model or learning architecture identified in brochure	Manufacturer algorithm description; not a comparative AI trial	V02; M06–M08	High for described mechanism; absence of ML documentation is not proof of no AI	Leadless pacing with automatic mechanical atrial sensing and AV-synchrony optimization
Micra VR2	Rate-response capability verified; AV2 AI label unsupported	Accelerometer-based rate response; mechanical atrial sensing listed N/A in p.4 specification table	Implanted ventricular pacemaker	No learned model identified; AV2 atrial-sensing algorithms should not be assigned to VR2	p.4 table checked as page image; not a patient-outcome evaluation	V02, p.4 (visual check)	High for feature distinction; limited for proprietary implementation	Rate-responsive leadless ventricular pacemaker; distinguish from AV2 AV-synchronous sensing
AVEIR DR i2i	Bidirectional electrical communication verified; AI/ML not established	RA and RV implants exchange conducted electrical signals beat-to-beat; programmable signal strength; wake-up and digital message protocol	Two implanted pacemakers; programming sessions customize communication	No training process or learned predictor disclosed in engineering explanation	Manufacturer chief-engineer interview, 14 December 2023	V03; M09–M10	High for vendor-described communication; not independent verification of all firmware	Dual-chamber leadless pacing coordinated through implant-to-implant electrical communication
EMBLEM S-ICD + EMPOWER	Commanded ATP and programmed sensing verified; AI/ML not established	S-ICD sends request; EMPOWER receives command and paces; EMPOWER has programmable fixed-threshold sensing and accelerometer rate response	Separate S-ICD and intracardiac EMPOWER	No learned model documented in examined protocol device/test sections; complete firmware not inspected	Sponsor protocol Version I, 29 April 2022; communication test is not efficacy result	V04; M11–M12; pp.23,124–125,130	High for specified command path; incomplete for every detection algorithm	Coordinated modular defibrillation and pacing system; AI contribution remains unverified
Qinming8631 DR	Rate-responsive pacing documented; detailed mechanism unresolved	Dual-chamber rate-responsive pacing tested; exact sensor and controller design not established by this trial	Implanted pacemaker; detailed algorithm unavailable	No trained-model architecture or training protocol in trial report	225 randomized; six-month pacing noninferiority trial, not AI validation	V05 (= original S03); M13	High for trial scope; low for internal controller classification	Dual-chamber rate-responsive pacemaker evaluated in a randomized noninferiority trial

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P01	S01	BIOMONITOR IV	§3.2 / §5.1, BIOTRONIK arrhythmia monitoring	References, asterisk	AI algorithm is applied only to AF detection	AI_EXPLICIT;SCOPE_LIMIT	qualifies	Explicit AI is narrower than all-arrhythmia or therapy claims.				BIOTRONIK announced first implant of new implantable cardiac monitor with artificial intelligence, targeting efficiency in monitoring for cardiac arrhythmias	BIOTRONIK	2023-06-12	Manufacturer press release	https://www.prnewswire.com/news-releases/biotronik-announced-first-implant-of-new-implantable-cardiac-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Explicit AI; AF detection only	Home Monitoring Service Center filtering setting; on-implant AI not established	Manufacturer; performance based on data on file	Read in live full-text web source; whitespace normalized only	PASS
P02	S01	BIOMONITOR IV	§3.2, AI integration	References, reference 1	With AF filtering criterion set to Very Specific in the Home Monitoring Service Center.	LOCUS;SCOPE_LIMIT	qualifies	Filtering setting is in the monitoring service; does not establish implant-resident learning.				BIOTRONIK announced first implant of new implantable cardiac monitor with artificial intelligence, targeting efficiency in monitoring for cardiac arrhythmias	BIOTRONIK	2023-06-12	Manufacturer press release	https://www.prnewswire.com/news-releases/biotronik-announced-first-implant-of-new-implantable-cardiac-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Explicit AI; AF detection only	Home Monitoring Service Center filtering setting; on-implant AI not established	Manufacturer; performance based on data on file	Read in live full-text web source; whitespace normalized only	PASS
P03	S01	BIOMONITOR IV	§3.2, diagnostic improvement	References, reference 1	Reduction is relative to performance without SmartECG.	DIAGNOSTIC;SCOPE_LIMIT	qualifies	Comparator is without SmartECG, not a head-to-head comparison across manufacturers.				BIOTRONIK announced first implant of new implantable cardiac monitor with artificial intelligence, targeting efficiency in monitoring for cardiac arrhythmias	BIOTRONIK	2023-06-12	Manufacturer press release	https://www.prnewswire.com/news-releases/biotronik-announced-first-implant-of-new-implantable-cardiac-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Explicit AI; AF detection only	Home Monitoring Service Center filtering setting; on-implant AI not established	Manufacturer; performance based on data on file	Read in live full-text web source; whitespace normalized only	PASS
P04	S01	BIOMONITOR IV	§3.2 / §8.1, monitoring accuracy	Main text, BIOMONITOR IV paragraph	to reduce all false positive detections by 86% while preserving 98% of true episodes	DIAGNOSTIC;QUANT_CLAIM	supports_noncausal	Manufacturer claim; reference is data on file; not a public clinical AI-effectiveness trial.	86% false-positive reduction; 98% episode preservation			BIOTRONIK announced first implant of new implantable cardiac monitor with artificial intelligence, targeting efficiency in monitoring for cardiac arrhythmias	BIOTRONIK	2023-06-12	Manufacturer press release	https://www.prnewswire.com/news-releases/biotronik-announced-first-implant-of-new-implantable-cardiac-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Explicit AI; AF detection only	Home Monitoring Service Center filtering setting; on-implant AI not established	Manufacturer; performance based on data on file	Read in live full-text web source; whitespace normalized only	PASS
P05	S02	Micra AV2 / Micra VR2	§4.4.1 / §4.4.4 / §5.3, battery longevity	Main text, battery paragraph	the median projected battery life of Micra AV2 and Micra VR2 is nearly 16 and 17 years, respectively.	LONGEVITY;PROJECTION	qualifies	Projected medians, not 16–17 years of observed follow-up; resolves conflicting numbers across the paper.	Nearly 16 / 17 years (projected)			Medtronic receives FDA approval for its next generation Micra leadless pacing systems	Medtronic plc	2023-05-01	Manufacturer press release	https://news.medtronic.com/2023-05-01-Medtronic-receives-FDA-approval-for-its-next-generation-Micra-leadless-pacing-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Automatic algorithms described; AI/ML not established	Implanted pacing algorithms (AV2); training method not reported	Manufacturer; longevity projection references internal modeling	Read in live full-text web source; whitespace normalized only	PASS
P06	S02	Micra AV2 / Micra VR2	§4.4.1 / §5.1, advanced AI algorithms	Main text, algorithms paragraph	The new Micra AV2 also includes advanced algorithms that automatically program AV synchrony, thereby coordinating the heart's upper and lower chambers.	AUTOMATION;THERAPY	supports_noncausal	Supports automatic AV synchrony in AV2; does not establish AI/ML or the same function in VR2.				Medtronic receives FDA approval for its next generation Micra leadless pacing systems	Medtronic plc	2023-05-01	Manufacturer press release	https://news.medtronic.com/2023-05-01-Medtronic-receives-FDA-approval-for-its-next-generation-Micra-leadless-pacing-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Automatic algorithms described; AI/ML not established	Implanted pacing algorithms (AV2); training method not reported	Manufacturer; longevity projection references internal modeling	Read in live full-text web source; whitespace normalized only	PASS
P07	S02	Micra AV2 / Micra VR2	§4.4, FDA approval	Main text, approval paragraph	approval of its Micra™ AV2 and Micra™ VR2	REGULATORY	supports_noncausal	Manufacturer reports FDA approval; not an independent FDA record or proof of AI status.				Medtronic receives FDA approval for its next generation Micra leadless pacing systems	Medtronic plc	2023-05-01	Manufacturer press release	https://news.medtronic.com/2023-05-01-Medtronic-receives-FDA-approval-for-its-next-generation-Micra-leadless-pacing-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Automatic algorithms described; AI/ML not established	Implanted pacing algorithms (AV2); training method not reported	Manufacturer; longevity projection references internal modeling	Read in live full-text web source; whitespace normalized only	PASS
P08	S03	Qinming8631 DR (QM7211 is ventricular lead)	§4.5, Qinming safety and efficacy	Methods > Study design	This prospective, multicenter, randomized controlled, single-blind, noninferiority study was conducted at 14 clinical sites in China between September 2013 and December 2014.	TRIAL_DESIGN	supports_noncausal	Primary comparative evidence, but not a trial isolating any AI contribution.		225 randomized (113 / 112)	6 months	Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients: A Prospective, Multicenter, Randomized Controlled Trial of the First Domestically Developed Pacemaker of China	Xiang, Mei-Xiang; Wang, Dong-Qi; Xu, Jing; et al.	2016-11-20	Original randomized clinical trial	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.	2026-10-28	Publisher marks article Open; copyright Chinese Medical Association; exact reuse licence not verified	Rate-responsive pacing; AI/ML not established	Pacemaker and leads; no learned model reported	Authors report no sponsorship and no conflicts; short follow-up	Read in live full-text web source; whitespace normalized only	MANUAL_WEB_FULLTEXT
P09	S03	Qinming8631 DR (QM7211 is ventricular lead)	§5.1–5.3, Qinming QM7211 device identification	Methods > Study design	QM7211 (Qinming Medical, Baoji, China) active lead as the ventricular electrode lead.	IDENTITY	contradicts	QM7211 is an electrode lead, not the dual-chamber pacemaker; model should be Qinming8631 DR.				Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients: A Prospective, Multicenter, Randomized Controlled Trial of the First Domestically Developed Pacemaker of China	Xiang, Mei-Xiang; Wang, Dong-Qi; Xu, Jing; et al.	2016-11-20	Original randomized clinical trial	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.	2026-10-28	Publisher marks article Open; copyright Chinese Medical Association; exact reuse licence not verified	Rate-responsive pacing; AI/ML not established	Pacemaker and leads; no learned model reported	Authors report no sponsorship and no conflicts; short follow-up	Read in live full-text web source; whitespace normalized only	MANUAL_WEB_FULLTEXT
P10	S03	Qinming8631 DR (QM7211 is ventricular lead)	§4.5.3, comparative performance	Abstract > Conclusions	This study established the noninferiority of Qinming8631 DR to Talos DR.	CLINICAL_ENDPOINT;TRIAL_DESIGN	supports_noncausal	Noninferiority of pacing is not evidence of superiority or AI integration.		225 randomized (113 / 112)	6 months	Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients: A Prospective, Multicenter, Randomized Controlled Trial of the First Domestically Developed Pacemaker of China	Xiang, Mei-Xiang; Wang, Dong-Qi; Xu, Jing; et al.	2016-11-20	Original randomized clinical trial	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.	2026-10-28	Publisher marks article Open; copyright Chinese Medical Association; exact reuse licence not verified	Rate-responsive pacing; AI/ML not established	Pacemaker and leads; no learned model reported	Authors report no sponsorship and no conflicts; short follow-up	Read in live full-text web source; whitespace normalized only	MANUAL_WEB_FULLTEXT
P11	S03	Qinming8631 DR (QM7211 is ventricular lead)	§4.5.3 / §7.1, long-term efficacy	Discussion > Limitations	The follow-up period in this study was relatively short according to the battery expectancy of a dual-chamber pacemaker.	SCOPE_LIMIT;LONGEVITY	qualifies	Long-term durability remains untested by this six-month study.			6 months	Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients: A Prospective, Multicenter, Randomized Controlled Trial of the First Domestically Developed Pacemaker of China	Xiang, Mei-Xiang; Wang, Dong-Qi; Xu, Jing; et al.	2016-11-20	Original randomized clinical trial	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.	2026-10-28	Publisher marks article Open; copyright Chinese Medical Association; exact reuse licence not verified	Rate-responsive pacing; AI/ML not established	Pacemaker and leads; no learned model reported	Authors report no sponsorship and no conflicts; short follow-up	Read in live full-text web source; whitespace normalized only	MANUAL_WEB_FULLTEXT
P12	S04	EMBLEM MRI S-ICD + EMPOWER MPS	§4.6, applications and benefits	Main text, trial design	The prospective, non-randomized MODULAR ATP clinical trial will enroll at up to 50 centers in the U.S., Canada and Europe up to 300 patients	TRIAL_DESIGN;ADOPTION_STAGE	qualifies	Planned enrollment, not completed results; do not combine with later trial findings.	Up to 50 centers	Up to 300 planned		Boston Scientific Initiates Trial to Evaluate Industry's First Modular CRM System	Boston Scientific Corporation	2021-12-02	Manufacturer trial-initiation announcement	https://news.bostonscientific.com/2021-12-02-Boston-Scientific-Initiates-Trial-to-Evaluate-Industrys-First-Modular-CRM-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Device coordination described; AI/ML not established	Communication between S-ICD and leadless pacemaker	Sponsor; prospective plans, not completed clinical results	Read in live full-text web source; whitespace normalized only	PASS
P13	S04	EMBLEM MRI S-ICD + EMPOWER MPS	§4.6, adoption and availability	Investigational-use footnote	The EMPOWER MPS is an investigational device and limited by U.S. law to investigational use only. It is not available for sale.	REGULATORY;ADOPTION_STAGE	qualifies	Status at source date, December 2021; does not independently establish February 2024 status.				Boston Scientific Initiates Trial to Evaluate Industry's First Modular CRM System	Boston Scientific Corporation	2021-12-02	Manufacturer trial-initiation announcement	https://news.bostonscientific.com/2021-12-02-Boston-Scientific-Initiates-Trial-to-Evaluate-Industrys-First-Modular-CRM-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Device coordination described; AI/ML not established	Communication between S-ICD and leadless pacemaker	Sponsor; prospective plans, not completed clinical results	Read in live full-text web source; whitespace normalized only	PASS
P14	S04	EMBLEM MRI S-ICD + EMPOWER MPS	§4.6.3 / §5.3, remains outside and never touches heart	Main text, S-ICD paragraph	because it provides protection without touching the heart or substernal space	IDENTITY;SCOPE_LIMIT	qualifies	This description applies to the S-ICD component, not the whole system including an intracardiac pacemaker.				Boston Scientific Initiates Trial to Evaluate Industry's First Modular CRM System	Boston Scientific Corporation	2021-12-02	Manufacturer trial-initiation announcement	https://news.bostonscientific.com/2021-12-02-Boston-Scientific-Initiates-Trial-to-Evaluate-Industrys-First-Modular-CRM-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Device coordination described; AI/ML not established	Communication between S-ICD and leadless pacemaker	Sponsor; prospective plans, not completed clinical results	Read in live full-text web source; whitespace normalized only	PASS
P15	S04	EMBLEM MRI S-ICD + EMPOWER MPS	§4.6 / §6.1, coordinated therapy	Main text, trial endpoints	the communication success between the EMPOWER MPS and the S-ICD System	INTEROPERABILITY; CLINICAL_ENDPOINT	supports_noncausal	Device-to-device communication is a trial endpoint; communication alone is not AI.				Boston Scientific Initiates Trial to Evaluate Industry's First Modular CRM System	Boston Scientific Corporation	2021-12-02	Manufacturer trial-initiation announcement	https://news.bostonscientific.com/2021-12-02-Boston-Scientific-Initiates-Trial-to-Evaluate-Industrys-First-Modular-CRM-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Device coordination described; AI/ML not established	Communication between S-ICD and leadless pacemaker	Sponsor; prospective plans, not completed clinical results	Read in live full-text web source; whitespace normalized only	PASS
P16	S05	AVEIR DR i2i	§4.7.2, AVEIR study sample	Late-Breaking Data section	primary safety and efficacy endpoints were analyzed in the first 300 people enrolled across 55 centers	TRIAL_DESIGN	supports_noncausal	Sponsor-reported early cohort, not a randomized comparison.	55 centers	300	3 months	Abbott Unveils Late-Breaking Data Showing the World's First Dual-Chamber Leadless Pacemaker Study Met All Three Primary Endpoints	Abbott	2023-05-20	Sponsor report of clinical trial results	https://abbott.mediaroom.com/2023-05-20-Abbott-Unveils-Late-Breaking-Data-Showing-the-Worlds-First-Dual-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Implant-to-implant communication; AI/ML not established	Two leadless pacemakers communicating beat-to-beat	Sponsor summary, not independently reanalyzed patient data	Read in live full-text web source; whitespace normalized only	PASS
P17	S05	AVEIR DR i2i	§4.7.2, implant success	Results bullet 1	Physicians demonstrated a 98.3% implant success rate.	CLINICAL_ENDPOINT;QUANT_CLAIM	supports_noncausal	Implantation endpoint; cannot attribute this percentage to AI.	98.3% implant success	300	3 months	Abbott Unveils Late-Breaking Data Showing the World's First Dual-Chamber Leadless Pacemaker Study Met All Three Primary Endpoints	Abbott	2023-05-20	Sponsor report of clinical trial results	https://abbott.mediaroom.com/2023-05-20-Abbott-Unveils-Late-Breaking-Data-Showing-the-Worlds-First-Dual-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Implant-to-implant communication; AI/ML not established	Two leadless pacemakers communicating beat-to-beat	Sponsor summary, not independently reanalyzed patient data	Read in live full-text web source; whitespace normalized only	PASS
P18	S05	AVEIR DR i2i	§4.7.2, AV synchrony	Results bullet 3	More than 97% of people had successful atrio-ventricular (AV) synchrony	CLINICAL_ENDPOINT;QUANT_CLAIM	supports_noncausal	People meeting synchrony endpoint; not interchangeable with average AV synchrony across postures.	>97% of people	300	3 months	Abbott Unveils Late-Breaking Data Showing the World's First Dual-Chamber Leadless Pacemaker Study Met All Three Primary Endpoints	Abbott	2023-05-20	Sponsor report of clinical trial results	https://abbott.mediaroom.com/2023-05-20-Abbott-Unveils-Late-Breaking-Data-Showing-the-Worlds-First-Dual-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Implant-to-implant communication; AI/ML not established	Two leadless pacemakers communicating beat-to-beat	Sponsor summary, not independently reanalyzed patient data	Read in live full-text web source; whitespace normalized only	PASS
P19	S05	AVEIR DR i2i	§4.7 / §8.2, AI applications	About the AVEIR DR i2i IDE Study	The AVEIR DR i2i IDE study is a prospective, multi-center, international, single-arm, pivotal investigational study.	TRIAL_DESIGN;SCOPE_LIMIT	qualifies	Single-arm design cannot estimate benefit over a concurrent control; AI is not specified.				Abbott Unveils Late-Breaking Data Showing the World's First Dual-Chamber Leadless Pacemaker Study Met All Three Primary Endpoints	Abbott	2023-05-20	Sponsor report of clinical trial results	https://abbott.mediaroom.com/2023-05-20-Abbott-Unveils-Late-Breaking-Data-Showing-the-Worlds-First-Dual-	2026-10-28	Publicly readable; copyrighted; no open reuse licence verified	Implant-to-implant communication; AI/ML not established	Two leadless pacemakers communicating beat-to-beat	Sponsor summary, not independently reanalyzed patient data	Read in live full-text web source; whitespace normalized only	PASS
P20	S06	Qualifying cyber devices (cross-cutting)	§6.1, security and regulation	§360n–2(b)(1)	a plan to monitor, identify, and address, as appropriate, in a reasonable time, postmarket cybersecurity vulnerabilities and exploits	CYBERSECURITY;REGULATORY	supports_context	Concrete lifecycle security obligation for submissions for qualifying cyber devices.				21 U.S.C. § 360n–2: Ensuring cybersecurity of devices (FD&C Act § 524B)	United States Congress	2022-12-29	Primary statute, reproduced by Cornell LII	https://www.law.cornell.edu/uscode/text/21/360n-2	2026-10-28	Underlying US statutory text public domain; LII editorial material not reproduced	Not AI-specific	Device and related systems; statutory definition controls applicability	Legal obligation, not evidence of clinical effectiveness	Read in live full-text web source; whitespace normalized only	PASS
P21	S06	Qualifying cyber devices (cross-cutting)	§6.1, security and regulation	§360n–2(b)(3)	a software bill of materials, including commercial, open-source, and off-the-shelf software components	CYBERSECURITY;REGULATORY	supports_context	Specific documentary requirement; not evidence that any named device complies.				21 U.S.C. § 360n–2: Ensuring cybersecurity of devices (FD&C Act § 524B)	United States Congress	2022-12-29	Primary statute, reproduced by Cornell LII	https://www.law.cornell.edu/uscode/text/21/360n-2	2026-10-28	Underlying US statutory text public domain; LII editorial material not reproduced	Not AI-specific	Device and related systems; statutory definition controls applicability	Legal obligation, not evidence of clinical effectiveness	Read in live full-text web source; whitespace normalized only	PASS
P22	S06	Qualifying cyber devices (cross-cutting)	§6.1, scope of cybersecurity obligations	§360n–2(c)(2)	has the ability to connect to the internet	SCOPE_LIMIT;CYBERSECURITY	qualifies	One of three conjunctive cyber-device criteria; do not assume every implant is covered.				21 U.S.C. § 360n–2: Ensuring cybersecurity of devices (FD&C Act § 524B)	United States Congress	2022-12-29	Primary statute, reproduced by Cornell LII	https://www.law.cornell.edu/uscode/text/21/360n-2	2026-10-28	Underlying US statutory text public domain; LII editorial material not reproduced	Not AI-specific	Device and related systems; statutory definition controls applicability	Legal obligation, not evidence of clinical effectiveness	Read in live full-text web source; whitespace normalized only	PASS

source_id	creator	date	title	url	device	genre	rights	relation	ai_status	locus	bias	accessed_date	retrieved_url	retrieved_html_sha256	access_note
S01	BIOTRONIK	2023-06-12	BIOTRONIK announced first implant of new implantable cardiac monitor with artificial intelligence, targeting efficiency in monitoring for cardiac arrhythmias	https://www.prnewswire.com/news-releases/biotronik-announced-first-implant-of-new-implantable-cardiac-monitor-with-artificial-intelligence-targeting-efficiency-in-monitoring-for-cardiac-arrhythmias-301848638.html	BIOMONITOR IV	Manufacturer press release	Publicly readable; copyrighted; no open reuse licence verified	Cited in manuscript pp. 15–16	Explicit AI; AF detection only	Home Monitoring Service Center filtering setting; on-implant AI not established	Manufacturer; performance based on data on file	2026-10-28	https://www.prnewswire.com/news-releases/biotronik-announced-first-implant-of-new-implantable-cardiac-monitor-with-artificial-intelligence-targeting-efficiency-in-monitoring-for-cardiac-arrhythmias-301848638.html	0c27c55a0b59fc5ce171332f9381c5742fd19a829b20583dde646d4d12555f5b	Full text read with web_fetch; automated re-fetch succeeded
S02	Medtronic plc	2023-05-01	Medtronic receives FDA approval for its next generation Micra leadless pacing systems	https://news.medtronic.com/2023-05-01-Medtronic-receives-FDA-approval-for-its-next-generation-Micra-leadless-pacing-systems	Micra AV2 / Micra VR2	Manufacturer press release	Publicly readable; copyrighted; no open reuse licence verified	Original manufacturer release behind manuscript's Wainscot Media citation	Automatic algorithms described; AI/ML not established	Implanted pacing algorithms (AV2); training method not reported	Manufacturer; longevity projection references internal modeling	2026-10-28	https://news.medtronic.com/2023-05-01-Medtronic-receives-FDA-approval-for-its-next-generation-Micra-leadless-pacing-systems	9b678e4338aa6ab5e25bf4e53573b03e93eaf2c2bb47327cfd2ee10c1ad6ae37	Full text read with web_fetch; automated re-fetch succeeded
S03	Xiang, Mei-Xiang; Wang, Dong-Qi; Xu, Jing; et al.	2016-11-20	Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients: A Prospective, Multicenter, Randomized Controlled Trial of the First Domestically Developed Pacemaker of China	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx	Qinming8631 DR (QM7211 is ventricular lead)	Original randomized clinical trial	Publisher marks article Open; copyright Chinese Medical Association; exact reuse licence not verified	Xiang et al. cited in manuscript p. 18; corrected DOI 10.4103/0366-6999.193443; PMCID PMC5126155	Rate-responsive pacing; AI/ML not established	Pacemaker and leads; no learned model reported	Authors report no sponsorship and no conflicts; short follow-up	2026-10-28	403 Client Error: Forbidden for url: https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx		Full text read with web_fetch; automated re-fetch unavailable
S04	Boston Scientific Corporation	2021-12-02	Boston Scientific Initiates Trial to Evaluate Industry's First Modular CRM System	https://news.bostonscientific.com/2021-12-02-Boston-Scientific-Initiates-Trial-to-Evaluate-Industrys-First-Modular-CRM-System	EMBLEM MRI S-ICD + EMPOWER MPS	Manufacturer trial-initiation announcement	Publicly readable; copyrighted; no open reuse licence verified	Primary sponsor source replacing manuscript's Prutchi blog report	Device coordination described; AI/ML not established	Communication between S-ICD and leadless pacemaker	Sponsor; prospective plans, not completed clinical results	2026-10-28	https://news.bostonscientific.com/2021-12-02-Boston-Scientific-Initiates-Trial-to-Evaluate-Industrys-First-Modular-CRM-System	90f32d8ee1e6dd90ec20b411dfdc2bd0fe2ff778b4bee9de742f390cfaf04fd4	Full text read with web_fetch; automated re-fetch succeeded
S05	Abbott	2023-05-20	Abbott Unveils Late-Breaking Data Showing the World's First Dual-Chamber Leadless Pacemaker Study Met All Three Primary Endpoints	https://abbott.mediaroom.com/2023-05-20-Abbott-Unveils-Late-Breaking-Data-Showing-the-Worlds-First-Dual-Chamber-Leadless-Pacemaker-Study-Met-All-Three-Primary-Endpoints	AVEIR DR i2i	Sponsor report of clinical trial results	Publicly readable; copyrighted; no open reuse licence verified	Directly cited in manuscript p. 15; figures reproduced pp. 8–9	Implant-to-implant communication; AI/ML not established	Two leadless pacemakers communicating beat-to-beat	Sponsor summary, not independently reanalyzed patient data	2026-10-28	https://abbott.mediaroom.com/2023-05-20-Abbott-Unveils-Late-Breaking-Data-Showing-the-Worlds-First-Dual-Chamber-Leadless-Pacemaker-Study-Met-All-Three-Primary-Endpoints	f178fad9ceff9457d76b4e1c4fb86f3a85893338548f02542f6a124a5f39ddb2	Full text read with web_fetch; automated re-fetch succeeded
S06	United States Congress	2022-12-29	21 U.S.C. § 360n–2: Ensuring cybersecurity of devices (FD&C Act § 524B)	https://www.law.cornell.edu/uscode/text/21/360n-2	Qualifying cyber devices (cross-cutting)	Primary statute, reproduced by Cornell LII	Underlying US statutory text public domain; LII editorial material not reproduced	Added primary legal evidence for manuscript §6.1	Not AI-specific	Device and related systems; statutory definition controls applicability	Legal obligation, not evidence of clinical effectiveness	2026-10-28	https://www.law.cornell.edu/uscode/text/21/360n-2	4d4b760b327fd20f340fef041f2b589399cb91c1ae014755d1f410eafb2512f3	Full text read with web_fetch; automated re-fetch succeeded

code	definition	paper_derivation
AI_EXPLICIT	Source expressly calls a function AI or ML; do not infer from automatic or adaptive.	§3.2; §8.2
AUTOMATION	Automatic algorithmic behavior without established learned-model provenance.	§4.4; §5.1
LOCUS	Where computation/filtering is described; unknown if not directly specified.	§3.1–3.2
DIAGNOSTIC	Arrhythmia detection/filtering, not therapy or patient outcome.	§3.2; §8.1
THERAPY	Direct pacing or treatment coordination.	§4.4–4.7
LONGEVITY	Battery/service life or need for long-term follow-up.	§4.4; §7.1
PROJECTION	Modeled/expected future outcome, not observed follow-up.	§4.4; §5.3
TRIAL_DESIGN	Design, comparator, enrollment or follow-up supporting inference limits.	§4.5; §4.7
CLINICAL_ENDPOINT	Reported clinical/device endpoint or prespecified endpoint, not AI attribution.	§4; §5
QUANT_CLAIM	Numerical performance assertion; retain denominator and evidence type.	§4; §5
IDENTITY	Correct device/model/component attribution.	§4.5–4.6; §5
INTEROPERABILITY	Device communication; not by itself EHR interoperability or AI.	§4.6; §6.1
ADOPTION_STAGE	Investigational, planned enrollment, or availability at source date.	§4; §6
REGULATORY	Approval report, investigational constraint or legal requirement.	§4; §6.1
CYBERSECURITY	Security obligations or scope of qualifying devices.	§6.1
SCOPE_LIMIT	Footnote, limitation, comparator or boundary narrowing an inference.	§5–7

passage_id	source_id	device	locator	quote	code	interpretation	creator	source_date	source_title	url	accessed_date	verification	rights
M01	V01	BIOMONITOR IV	Methods > SmartECG algorithm	SmartECG is a software package implemented in the HMSC for the BioMonitor IV ICM.	LOCUS_CONFIRMED	Confirms service-center computation, not implanted AI.	Bisignani et al.	2024-01-23	Implantable cardiac monitors: artificial intelligence and signal processing reduce remote ECG review workload and preserve arrhythmia detection sensitivity	https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2024.1343424/full	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	CC BY; publicly readable
M02	V01	BIOMONITOR IV	Methods > SmartECG algorithm	For the analysis of AF events only, it further uses artificial intelligence, which can be set to different levels of strictness, from very sensitive to very specific.	AI_EXPLICIT	AI limited to AF; remaining filtering includes signal processing.	Bisignani et al.	2024-01-23	Implantable cardiac monitors: artificial intelligence and signal processing reduce remote ECG review workload and preserve arrhythmia detection sensitivity	https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2024.1343424/full	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	CC BY; publicly readable
M03	V01	BIOMONITOR IV	Discussion > Remote vs. in-office follow-up	The SmartECG algorithm is implemented in the remote monitoring platform because it cannot be implemented in the ICM or in the programming device used to interrogate the ICM during a follow-up visit.	LOCUS_CONFIRMED	Directly rules out implant/programmer execution for this described SmartECG implementation.	Bisignani et al.	2024-01-23	Implantable cardiac monitors: artificial intelligence and signal processing reduce remote ECG review workload and preserve arrhythmia detection sensitivity	https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2024.1343424/full	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	CC BY; publicly readable
M04	V01	BIOMONITOR IV	Methods > Data sources and patient populations	patients who had contributed sECGs to the training of SmartECG artificial intelligence algorithm for atrial fibrillation (AF) recognition.	TRAINING_EVIDENCE	Excerpt from exclusion criterion: patient ECGs were used for training; excluded from evaluation. Architecture and complete training-set composition not disclosed.	Bisignani et al.	2024-01-23	Implantable cardiac monitors: artificial intelligence and signal processing reduce remote ECG review workload and preserve arrhythmia detection sensitivity	https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2024.1343424/full	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	CC BY; publicly readable
M05	V01	BIOMONITOR IV	Methods > SmartECG algorithm	Our analysis uses the nominal “balanced” setting.	VALIDATION_BOUNDARY	Not the Very Specific setting behind the 2023 marketing claim.	Bisignani et al.	2024-01-23	Implantable cardiac monitors: artificial intelligence and signal processing reduce remote ECG review workload and preserve arrhythmia detection sensitivity	https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2024.1343424/full	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	CC BY; publicly readable
M06	V02	Micra AV2	p. 12, Auto+ A3 Threshold	Auto+ uses filtered, true A3 signal amplitudes to automatically set the A3 Threshold above the A3 signal, but below the A7 signal (summated A3 + A4 signal)	CONTROL_RULE	Describes an adaptive threshold rule; no learned model specified.	Medtronic	2023-05	Micra leadless pacemakers; UC202002938e EN 05/2023	https://pdfs.semanticscholar.org/57d9/24ea3b36b976d9deeb0442f3d027ac70fc8d.pdf	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Copyright Medtronic; short attributed excerpts
M07	V02	Micra AV2	p. 13, Auto PVAB and Upper Tracking Rate	Micra AV2 offers an Auto PVAB algorithm which adjusts PVAB based on ventricular rate.	CONTROL_RULE	Rate-dependent blanking control, not evidence of ML.	Medtronic	2023-05	Micra leadless pacemakers; UC202002938e EN 05/2023	https://pdfs.semanticscholar.org/57d9/24ea3b36b976d9deeb0442f3d027ac70fc8d.pdf	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Copyright Medtronic; short attributed excerpts
M08	V02	Micra AV2	p. 11, Atrial Sensing Setup	Micra AV2 will automatically set up AV synchrony parameters Atrial Sensing Vector, A3 and A4 Threshold, and A3 Window End related parameters after implant.	AUTOMATION	Device-local parameter setup; automatic does not establish ML.	Medtronic	2023-05	Micra leadless pacemakers; UC202002938e EN 05/2023	https://pdfs.semanticscholar.org/57d9/24ea3b36b976d9deeb0442f3d027ac70fc8d.pdf	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Copyright Medtronic; short attributed excerpts
M09	V03	AVEIR DR i2i	Q: How do the leadless devices communicate?	Unlike traditional wireless methods, i2i utilizes conducted electrical signaling akin to electrical Morse code.	COMMUNICATION	Electrical communication mechanism, not a learning algorithm.	Abbott Cardiac Rhythm Management	2023-12-14	Unveiling the i2i Communication Technology in the AVEIR DR Leadless Pacemaker (LP) System	https://www.cardiovascular.abbott/us/en/hcp/products/cardiac-rhythm-management/blog/unveiling-the-i2i-communication-technology-in-aveir-dr.html	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Copyright Abbott; short attributed excerpts
M10	V03	AVEIR DR i2i	Q: What factors could impact i2i communication?	The strength of i2i signals can be adjusted based on measurements of i2i functionality and additional diagnostics obtained during programming sessions, providing customization for each patient.	PROGRAMMING	Clinical customization is documented; autonomous model training is not.	Abbott Cardiac Rhythm Management	2023-12-14	Unveiling the i2i Communication Technology in the AVEIR DR Leadless Pacemaker (LP) System	https://www.cardiovascular.abbott/us/en/hcp/products/cardiac-rhythm-management/blog/unveiling-the-i2i-communication-technology-in-aveir-dr.html	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Copyright Abbott; short attributed excerpts
M11	V04	EMBLEM + EMPOWER	Appendix A §26.1.1, p. 124	Fixed threshold sensing with programmable sense threshold	CONTROL_RULE	Explicitly describes fixed/programmed sensing for EMPOWER; do not extrapolate to every S-ICD detection algorithm.	Boston Scientific	2022-04-29	MODULAR ATP Clinical Investigation Plan, 92272548, Version I; NCT04798768	https://cdn.clinicaltrials.gov/large-docs/68/NCT04798768/Prot_000.pdf	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Public registry download; copyrighted; marked Confidential / do not copy or distribute; only short attributed excerpts retained
M12	V04	EMBLEM + EMPOWER	Appendix C §26.3, p. 130	Communication is tested by commanding a communication signal from the S-ICD System that, when successfully received by the EMPOWER PG, results in the EMPOWER PG delivering ATP therapy for up to 10 beats slightly faster than the subject's intrinsic rate.	COMMUNICATION	Command-to-therapy relationship documented in protocol test; no ML described.	Boston Scientific	2022-04-29	MODULAR ATP Clinical Investigation Plan, 92272548, Version I; NCT04798768	https://cdn.clinicaltrials.gov/large-docs/68/NCT04798768/Prot_000.pdf	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Public registry download; copyrighted; marked Confidential / do not copy or distribute; only short attributed excerpts retained
M13	V05	Qinming8631 DR	Abstract > Methods	Electrical properties, magnet response, single- and double-pole polarity conversion, rate response function, and adverse events of the pacing system were analyzed.	EVIDENCE_LIMIT	Trial assesses pacing functions but does not disclose a trained model or establish AI integration.	Xiang et al.	2016-11-20	Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx	2026-10-28	Full source read; HTML entities decoded / whitespace normalized; PDF line breaks joined	Publisher marked Open; exact reuse licence not verified

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V01	Bisignani et al.	2024-01-23	Implantable cardiac monitors: artificial intelligence and signal processing reduce remote ECG review workload and preserve arrhythmia detection sensitivity	https://www.frontiersin.org/journals/cardiovascular-medicine/articles/10.3389/fcvm.2024.1343424/full	Original pooled observational analysis	CC BY; publicly readable	Before nominal February 2024 publication, after manuscript's cited 16 January retrieval dates	Manufacturer-funded; Biotronik employee coauthor; BioMonitor IIIm recordings used to evaluate IV software workflow
V02	Medtronic	2023-05	Micra leadless pacemakers; UC202002938e EN 05/2023	https://pdfs.semanticscholar.org/57d9/24ea3b36b976d9deeb0442f3d027ac70fc8d.pdf	Manufacturer technical brochure, 20 pages	Copyright Medtronic; short attributed excerpts	Contemporaneous; dated May 2023 in footer	Manufacturer PDF read from Semantic Scholar mirror; official download returned HTTP 403; p.4 visually checked
V03	Abbott Cardiac Rhythm Management	2023-12-14	Unveiling the i2i Communication Technology in the AVEIR DR Leadless Pacemaker (LP) System	https://www.cardiovascular.abbott/us/en/hcp/products/cardiac-rhythm-management/blog/unveiling-the-i2i-communication-technology-in-aveir-dr.html	Manufacturer interview with chief engineer	Copyright Abbott; short attributed excerpts	Contemporaneous	Engineering explanation, not an independent efficacy study
V04	Boston Scientific	2022-04-29	MODULAR ATP Clinical Investigation Plan, 92272548, Version I; NCT04798768	https://cdn.clinicaltrials.gov/large-docs/68/NCT04798768/Prot_000.pdf	Publicly posted sponsor trial protocol, 148 pages	Public registry download; copyrighted; marked Confidential / do not copy or distribute; only short attributed excerpts retained	Document dated before paper; first public posting date not verified	Read device and communication sections, not every protocol page; not a source-code audit
V05	Xiang et al.	2016-11-20	Evaluation of Safety and Efficacy of Qinming8631 DR Implantable Cardiac Pacemaker in Chinese Patients	https://journals.lww.com/cmj/fulltext/2016/11200/evaluation_of_safety_and_efficacy_of_qinming8631.3.aspx	Original randomized trial; same document as S03	Publisher marked Open; exact reuse licence not verified	Contemporaneous	Rate response assessed; detailed sensor/controller design not specified; no AI attribution established